Gravitational waves sourced by magnetohydrodynamic turbulence in the early Universe

KAAI workshop Fall 2026

AI Methods for Turbulence and Early-Universe Signals

Pittsburgh (Nov. 2, 2026)

Alberto Roper Pol
University of Geneva

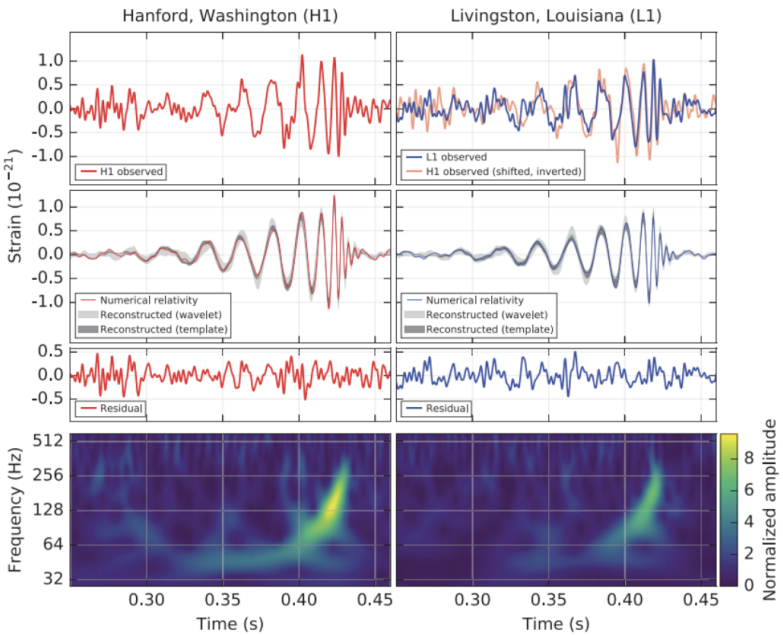
SNSF Ambizione grant (2023–2027): "*Exploring the early universe with gravitational waves and primordial magnetic fields.*"

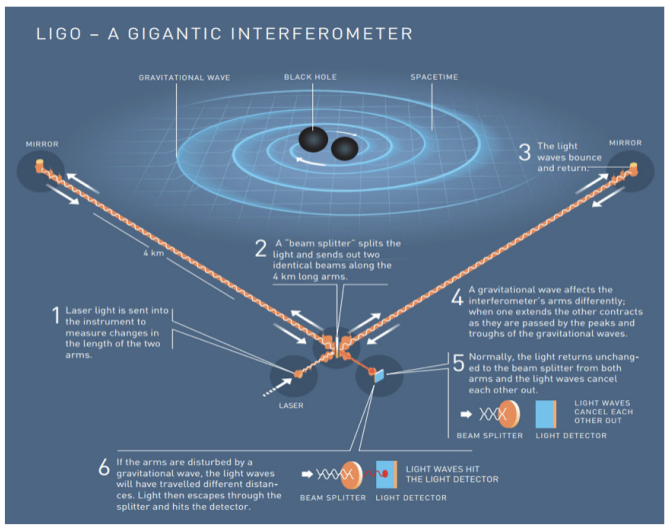
Keystone Astronomy & AI (KAAI) Fellow

McWilliams Center for Cosmology & Astrophysics

First direct detections of gravitational waves

- Gravitational waves are opening a new window into our understanding of the Universe
  - First event GW150914 detected by LIGO-Virgo collaboration[1]

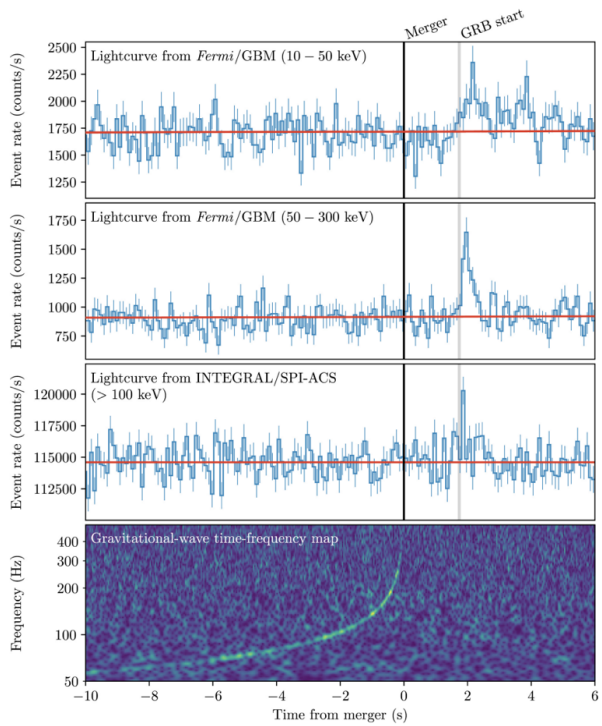


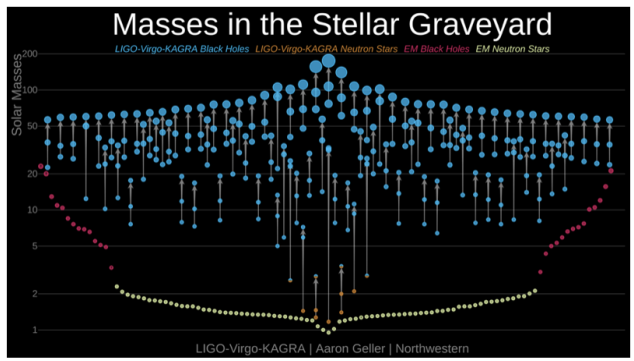


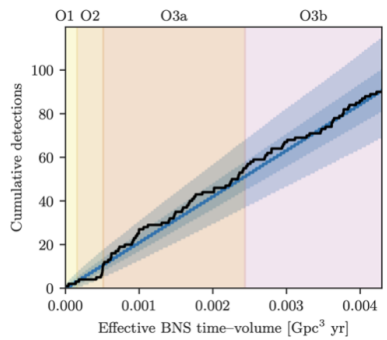
[1] [LIGO-Virgo Collaboration], *Phys. Rev. Lett.* **116**, 061102 (2016)

Introduction and Motivation

- GW170817 NS binary merger[2] first detection of GW and EM counterpart (constraint on the GW speed, measure of the Hubble rate, neutron star equation of state, ...)
- Several following events: LIGO-Virgo-KAGRA concluded the fourth observing run in November 2025 $\rightarrow \sim 340$ events up to O4 (90 up to O3b)[3] and 250 during O4.







[2] [LIGO-Virgo-Fermi GBM-Integral collaborations], Astrophys.J.Lett. 848 (2017) 2, L13

[3] [LIGO-Virgo Collaboration], GWTC-3, arXiv:2111.03606 (2021).

Cosmological Gravitational Waves as a test of BSM physics

- Gravity is the *weakest fundamental force*. Hence, GWs are difficult to detect but they propagate freely carrying *clean information of the source.*

- Cosmological GWs from the early Universe have the potential to provide us with *direct information on early universe physics* that is *not accessible via electromagnetic observations, complementary to collider experiments*.

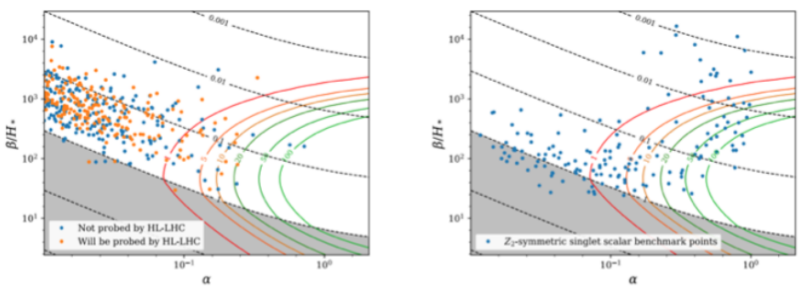
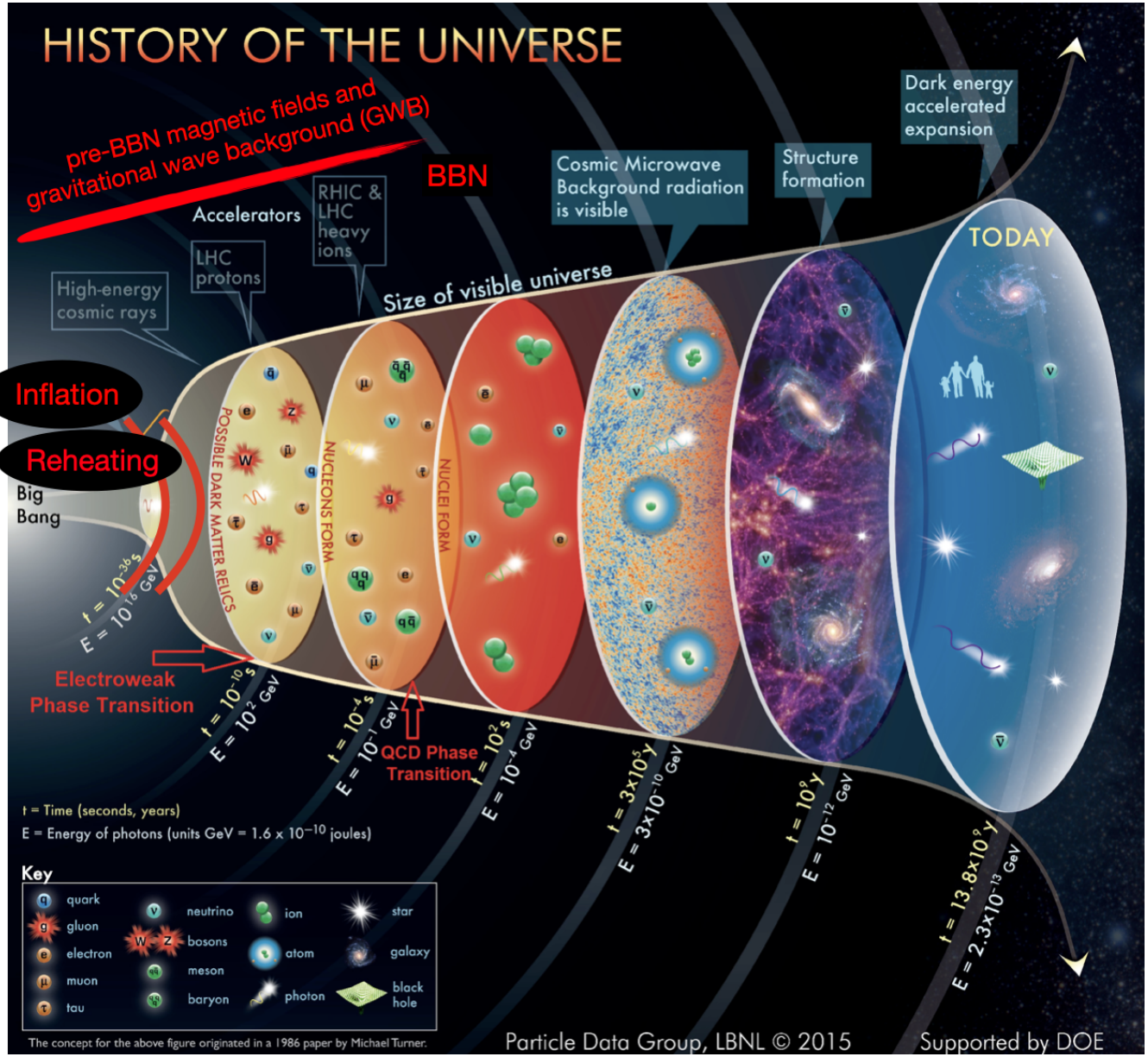


Figure 4: LEFT: Predicted values of α and β/H for $m_2 = 170$ GeV and 240 GeV (combined in one plot) in the general singlet model obtained by linearly varying the free parameters of the model and imposing requirements as described in the text. The mixing angles considered were $\sin\theta = 0.01$ (blue points) and 0.1 (orange points). The models in blue (orange) are unlikely (likely) to be probed by the high-luminosity LHC. The expected LISA sensitivities correspond to $T_* = 50$ GeV. RIGHT: Sensitivity curve for the Z_2 symmetric singlet extension. In both the left and right panels we have taken $v_w = 1$.

From [LISA CosWG], Caprini et al., *Detecting gravitational waves from cosmological phase transitions with LISA: an update* (2019)


HISTORY OF THE UNIVERSE
pre-BBN magnetic fields and gravitational wave background (GWB)
BBN
Accelerators
RHIC & LHC heavy ions
LHC protons
High-energy cosmic rays
Cosmic Microwave Background radiation is visible
Structure formation
Dark energy accelerated expansion
TODAY
Size of visible universe
Inflation
Reheating
Big Bang
POSSIBLE DARK MATTER RELICS
NUCLEONS FORM
NUCLEI FORM
t = 10^-35 s
E = 10^16 GeV
Electroweak Phase Transition
t = 10^-10 s
E = 10^2 GeV
t = 10^-4 s
E = 10^-1 GeV
QCD Phase Transition
t = 10^2 s
E = 10^-4 GeV
t = 3x10^5 y
E = 3x10^-10 GeV
t = 10^9 y
E = 10^-12 GeV
t = 13.8x10^9 y
E = 2.3x10^-13 GeV
t = Time (seconds, years)
E = Energy of photons (units GeV = 1.6 x 10^-10 joules)
Key
quark
gluon
electron
muon
tau
neutrino
bosons
meson
baryon
ion
atom
photon
star
galaxy
black hole
The concept for the above figure originated in a 1986 paper by Michael Turner.
Particle Data Group, LBNL © 2015
Supported by DOE

LISA

- Laser Interferometer Space Antenna (LISA) is a space–based GW detector
- Approved in 2017 as one of the main research missions of ESA (L3) with NASA collaboration
- Launch planned for 2035
- Composed by three spacecrafts in a distance of 2.5M km
- LISA cosmology working group (since 2015, $\sim$ 230 members)

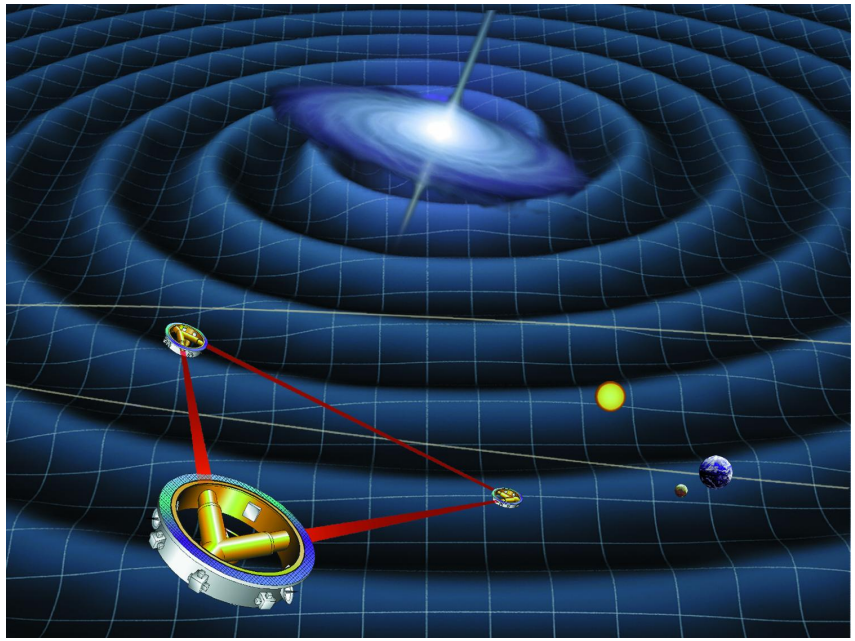
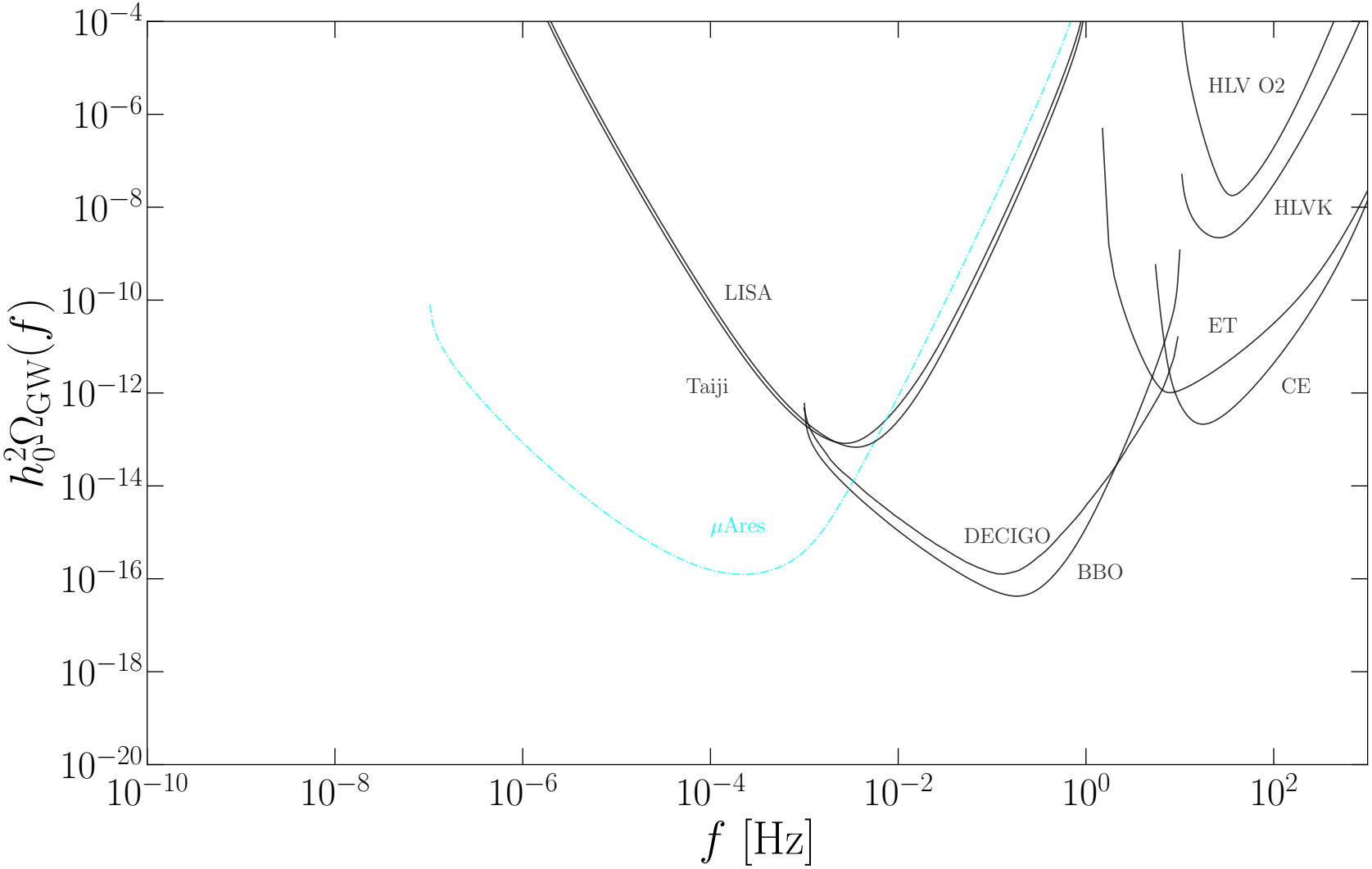


Figure: Artist's impression of LISA from Wikipedia

White paper:

[LISA Cosmology Working Group] (incl. ARP), *Living Rev. Rel.* **26** (2023), arXiv:2204.05434.

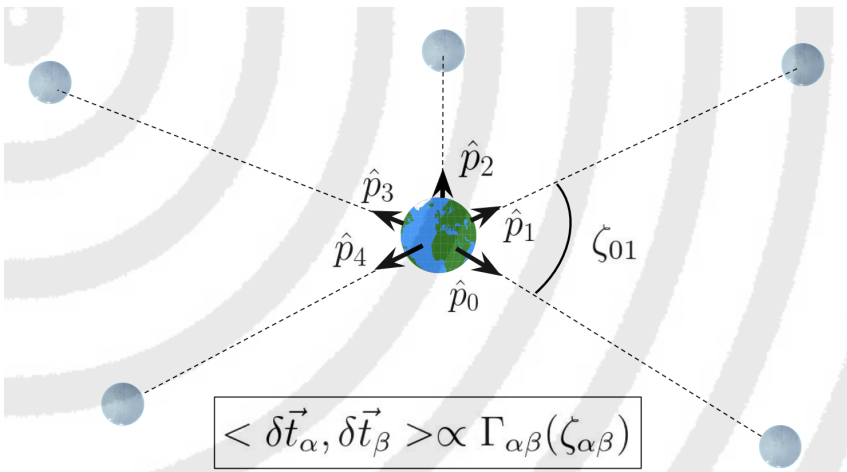
Gravitational spectrum (space-based detectors)



Pulsar Timing Array (PTA)

- An array of millisecond pulsars (MSP) is observed in the radio band to compute the delays on the time of arrival due to the presence of GWs.
- Collected data is the time series of residuals for each pulsar:

$$\delta t^i = t^i_{\rm obs} - t^i_{\rm TM}$$



Credit: Mikel Falxa

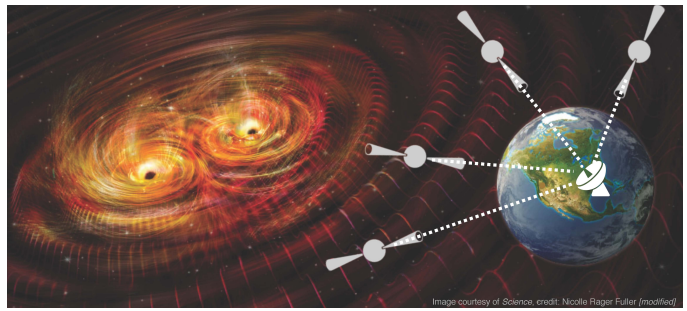
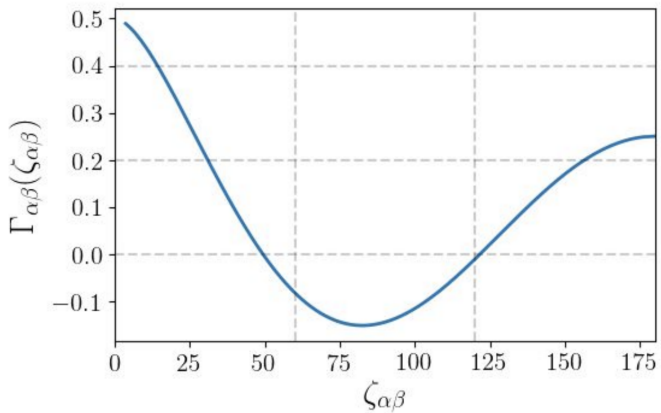


Figure: Image courtesy of Science, credit: Nicolle Rager Fuller

The correlation $\Gamma_{\alpha\beta}$ follows in GR the Hellings-Downs curve[4]

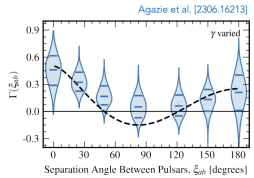


[4] R. W. Hellings and G. S. Downs, *Astrophys. J. Lett.* **265** (1983) L39-L42.

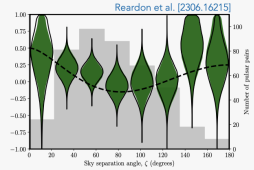
Pulsar Timing Array (PTA) "detection"

- The PTA collaborations reported for the first-time evidence of a stochastic gravitational wave background on a press release on June 28, 2023, followed by data release and collaboration publications.

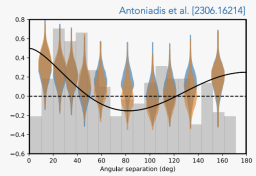
NANOGrav:
68 pulsars, 16yr of data
~3-4σ significance



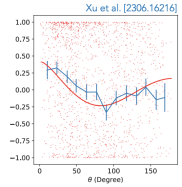
PPTA:
32 pulsars, 18yr of data
~2σ significance



EPTA + InPTA:
25 pulsars, 24yr of data
~3σ significance



CPTA:
57 pulsars, 3yr of data
~4.6σ significance



Credit: Andrea Mitridate

- Main candidate is the superposition of supermassive black hole binaries.
- However, the signal is compatible with "*more exciting*" sources of cosmological origin contributing to the background: cosmic strings, first-order phase transitions, primordial turbulence, primordial black holes, inflation. [5]

[5] [EPTA Collaboration] (incl. ARP), *The second data release from the European Pulsar Timing Array: V. Implications for massive black holes, dark matter and the early Universe,* arXiv:2306.16227.
[NANOGrav Collaboration], *The NANOGrav 15 yr Data Set: Search for Signals from New Physics,* arXiv:2306.16219

Probing the early Universe with GWs

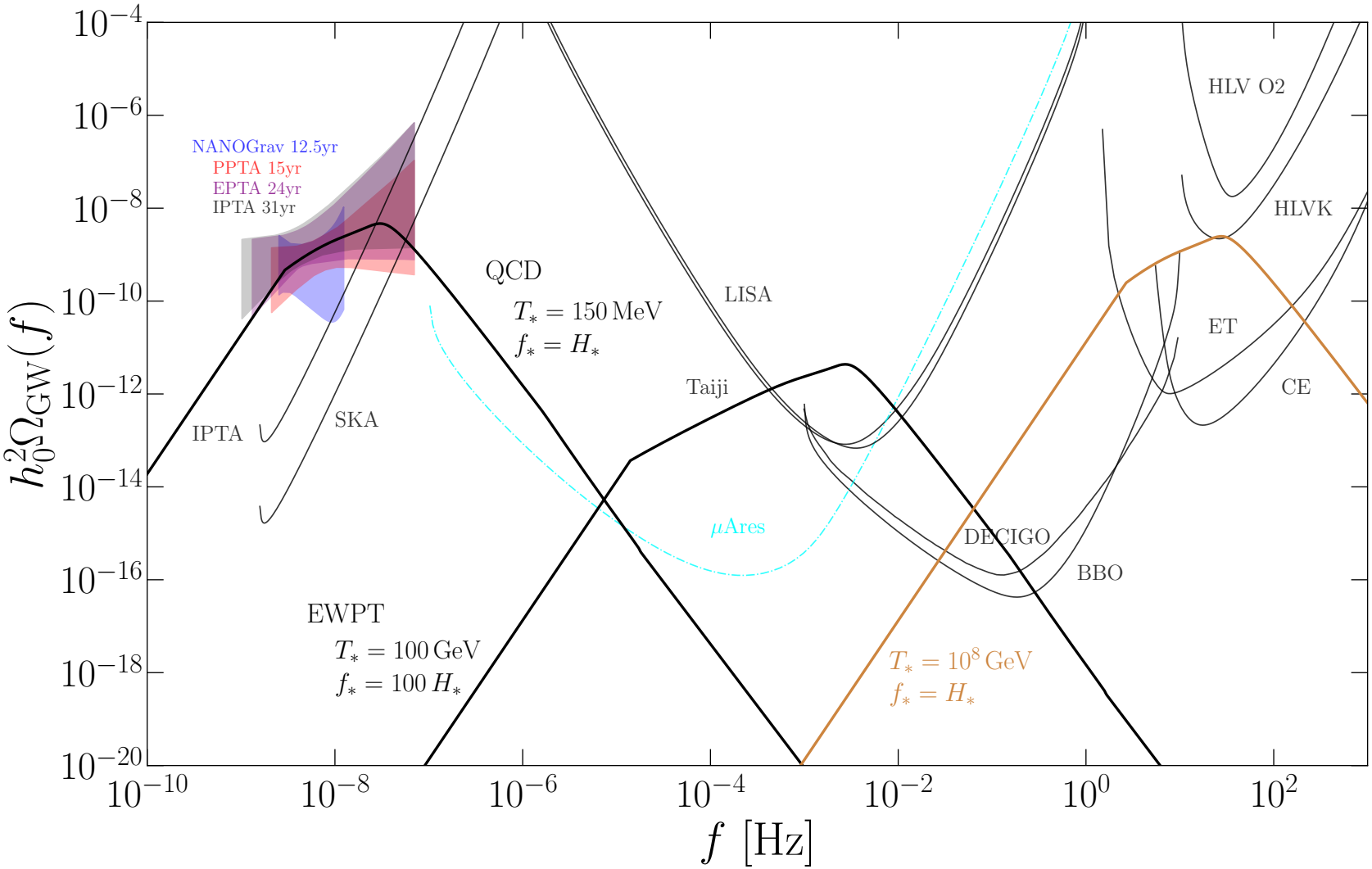
Cosmological (pre-recombination) GW background

- Why background? Individual sources are not resoluble, superposition of single events occurring in the whole Universe.

$$f_* \simeq 1.64 \times 10^{-3} \frac{0.01}{R_* \mathcal{H}_*} \frac{T_*}{100\,\mathrm{GeV}}\,\mathrm{Hz}$$

- Probing different energy scales
  - Ground-based detectors (LVK, ET, CE) sensitive to 10–1000 Hz
    Peccei-Quinn, B-L, left-right symmetries $\sim 10^7, 10^8$ GeV.
  - Space-based detectors **(LISA)** sensitive to 10^{-5}–10^{-2} Hz
    Electroweak phase transition $\sim$ 100 GeV
  - Pulsar Timing Array **(PTA)** sensitive to 10^{-9}–10^{-7} Hz
    Quark confinement (QCD) phase transition $\sim$ 100 MeV

Gravitational spectrum from decaying turbulence[6]

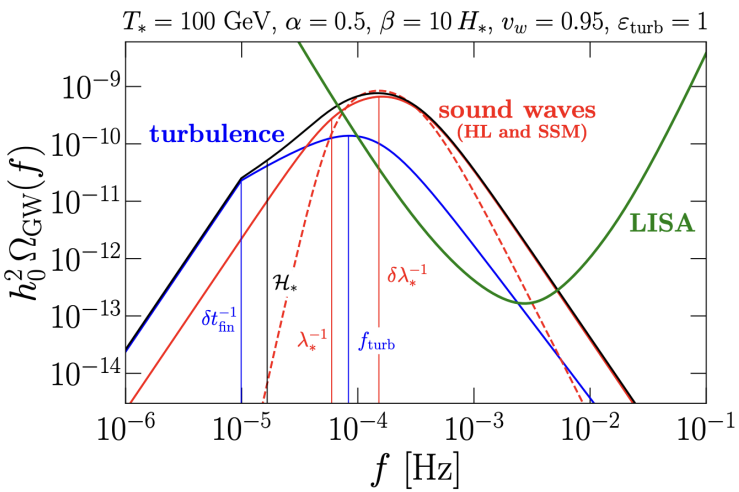


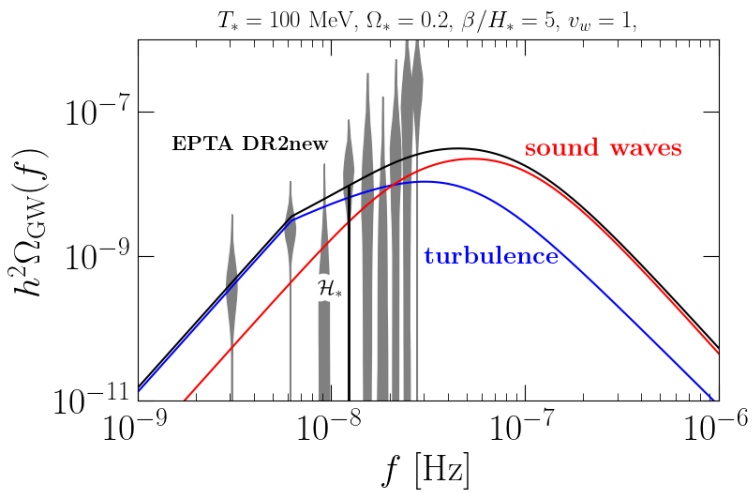
[6] ARP, C. Caprini, A. Neronov, D. Semikoz, *PRD* **105**, 123502 (2022), arXiv:2201.05630.
A. Neronov, ARP, C. Caprini, D. Semikoz, *PRD* **103**, L041302 (2021), arXiv:2009.14174.
ARP *et al.*, A&A **708** (2026) A337, arXiv:2307.10744.

MHD sources of GWs in the early Universe

- Magnetohydrodynamic (MHD) sources of GWs:
  - Sound waves generated from first-order phase transitions.
  - (M)HD turbulence in the primordial plasma.
  - Primordial magnetic fields.

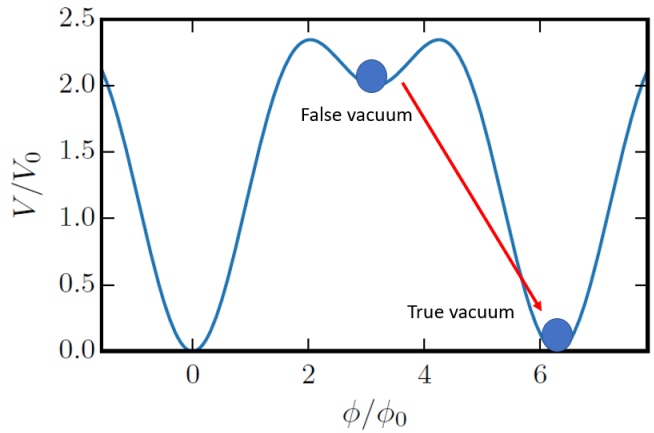
ARP *et al.*, 2307.10744, 2308.12943; [EPTA/InPTA] (incl. ARP), 2306.16227

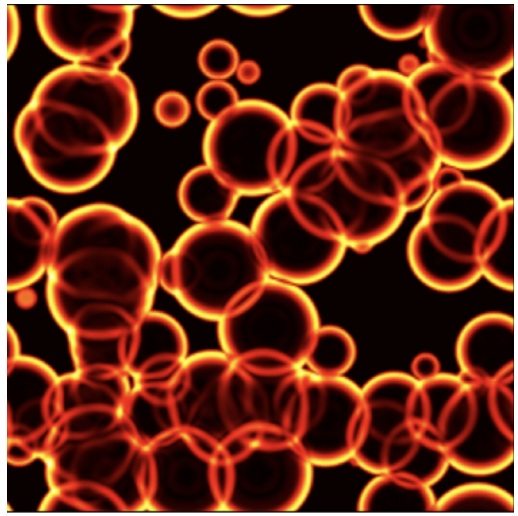




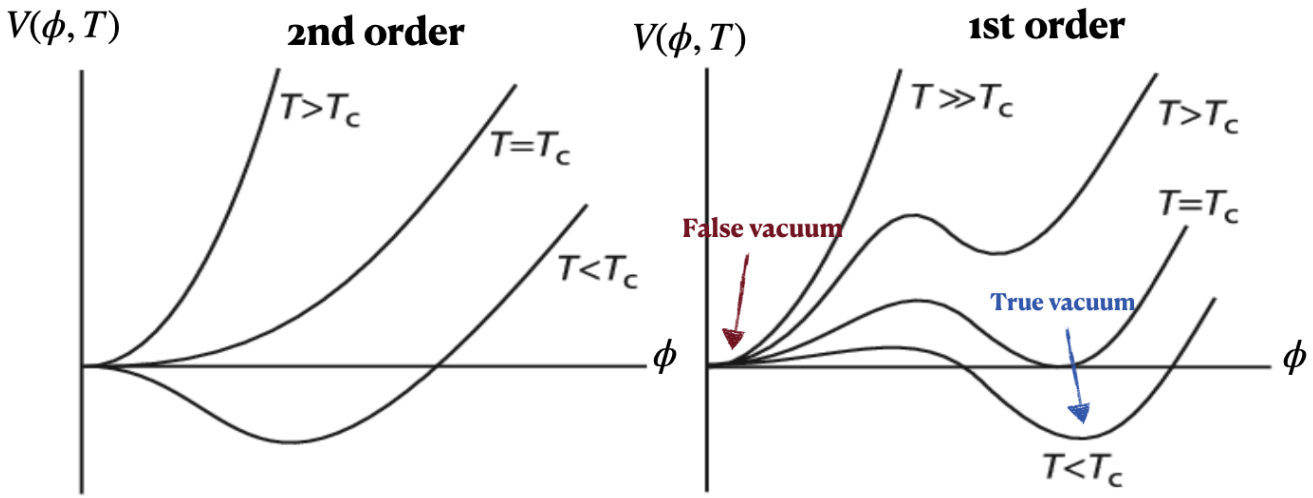
First–order phase transition

$$V(\phi, T) = \frac{1}{2}M^2(T)\phi^2 - \frac{1}{3}\delta(T)\phi^3 + \frac{1}{4}\lambda\phi^4$$



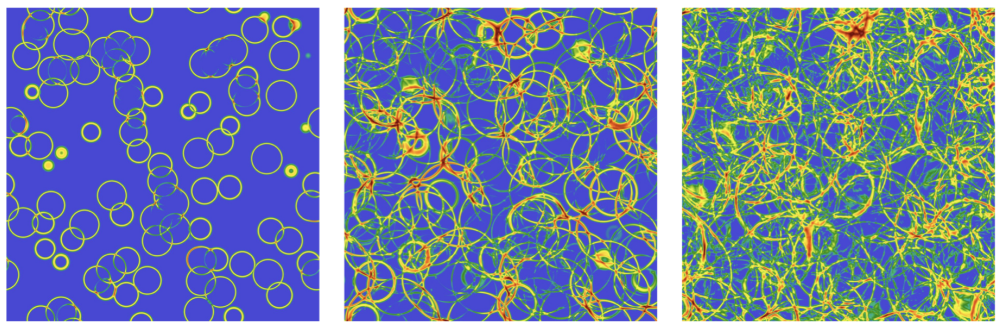


Credits: I. Stomberg

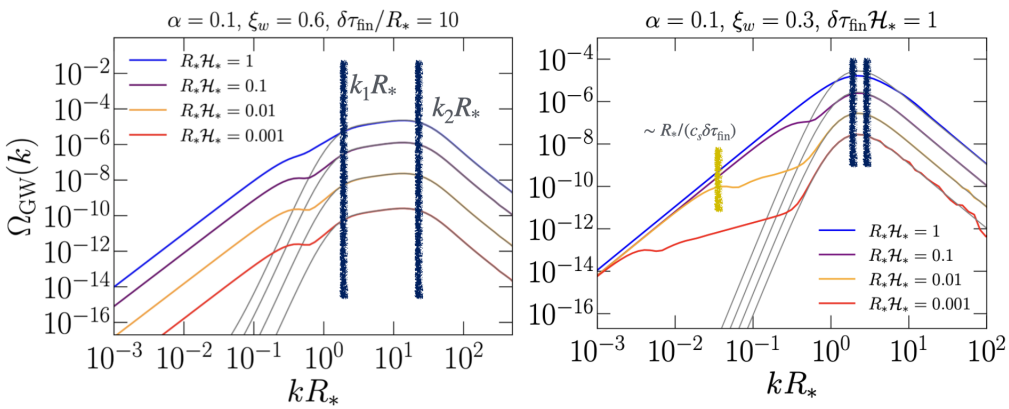


GWs from phase transitions

- Numerical simulations of the scalar + fluid system performed by the Sussex/Helsinki group via an effective friction term indicate sound-wave regime to dominate for weak/intermediate phase transitions.[7]



- Sound-shell model.[8] Three scales are found to determine the GW spectrum: $k_1 = R_*^{-1}$, $k_2 = \Delta R_*^{-1}$ (sound-shell thickness), and $\delta\tau_{\rm fin}$ (sound-wave regime duration).



$$\Omega_{\rm GW}(k) = 3\,\tilde{\Omega}_{\rm GW}\,K^2\,\Upsilon(\tau_{\rm fin})\,(H_*R_*)\,S(kR_*)$$

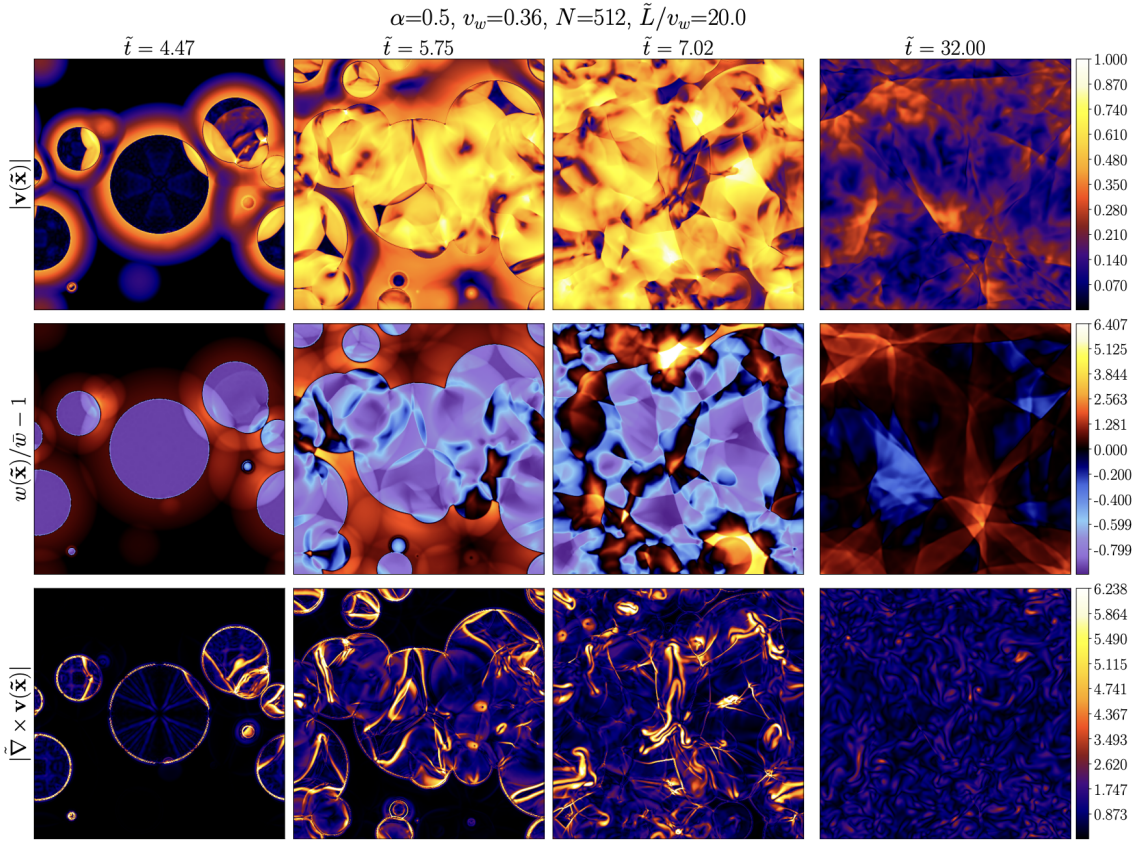
$$\Upsilon(\tau_{\rm fin}) = \frac{H_*\tau_{\rm fin}}{1 + H_*\tau_{\rm fin}}$$

$$S(kR_*) = \mathcal{N}\left(\frac{k}{k_1}\right)^{n_1}\left[1 + \left(\frac{k}{k_1}\right)^{a_1}\right]^{\frac{-n_1+n_2}{a_1}}\left[1 + \left(\frac{k}{k_2}\right)^{a_2}\right]^{\frac{-n_2+n_3}{a_2}}$$

[7] Hindmarsh *et al.*, 2013, 2015, 2017, Cutting *et al.*, 2019, Correia *et al.*, 2025.

[8] Hindmarsh, 2016, Hindmarsh, Hijazi, 2019, ARP, Procacci, Caprini, 2023.

Higgsless simulations of strong PTs[9]



Chiara Caprini
Professor at University of Geneva and staff member at CERN
Website

Isak Stomberg
Postdoc at IFIC, Valencia
Website

Thomas Konstandin
Scientist at DESY, Hamburg
Website

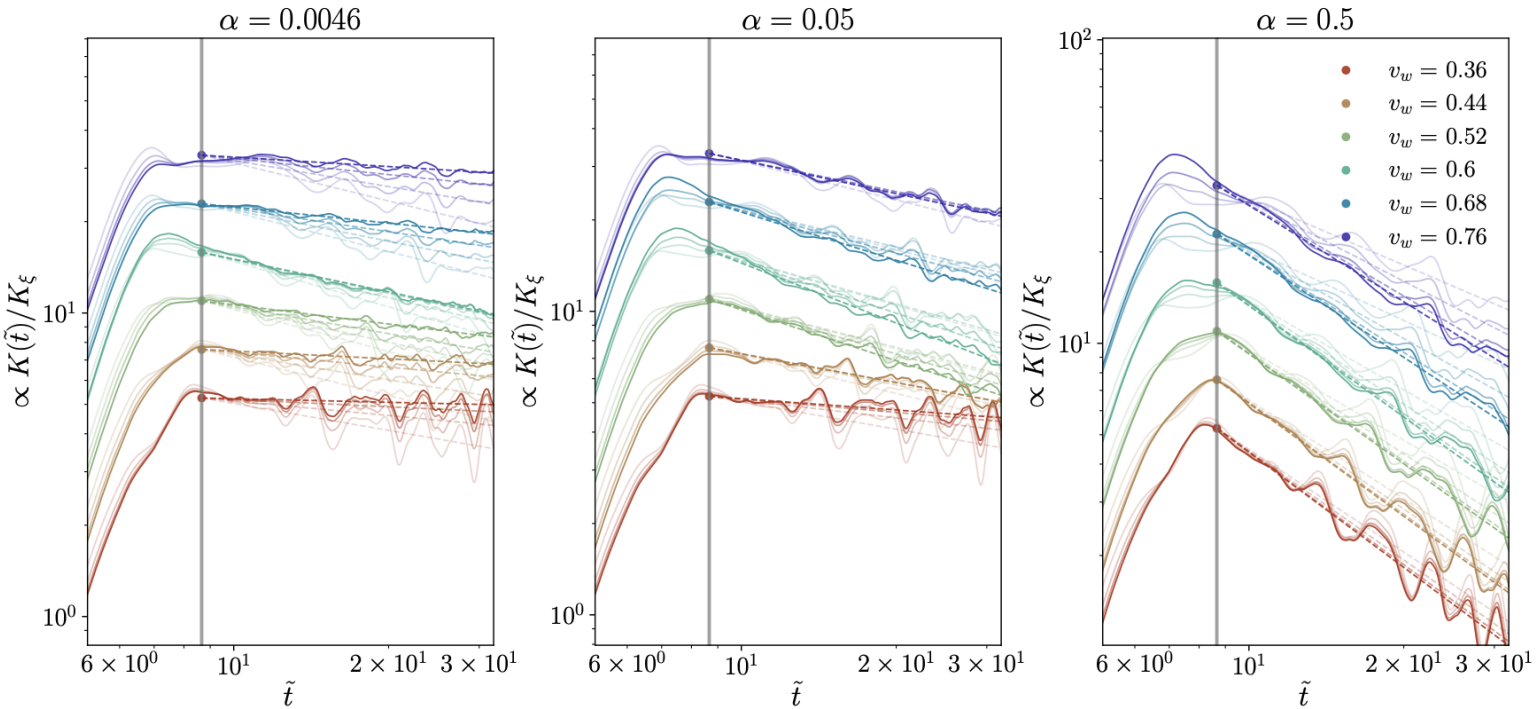
Ryusuke Jinno
Professor at Kobe University
Website

Henrique Rubira
Postdoc at LMU and Cambridge
Website

[9] ARP, Stomberg et al., JHEP **07** (2025) 217, arXiv:2409.03651.

Higgsless simulations (results)[10]

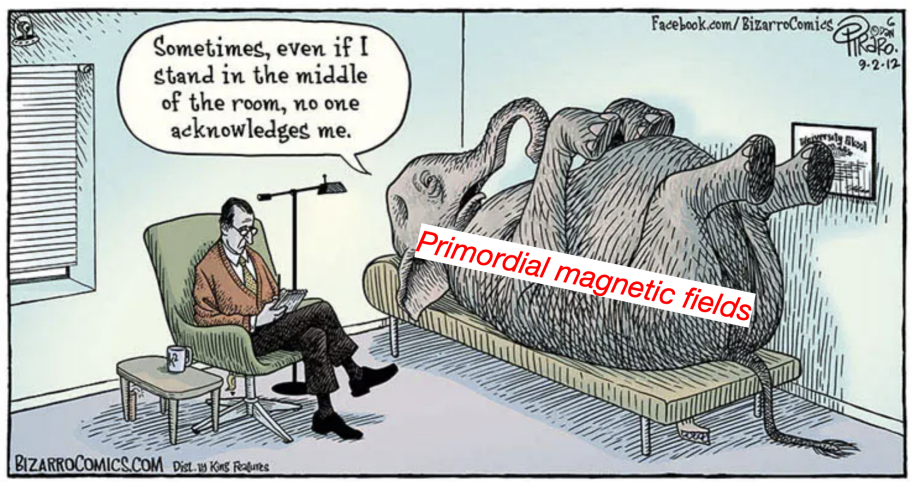
- Kinetic energy decay is observed in the simulations.
- For weak and strong PTs, increasing discretization enhances the decay.
- Potential indication of the development of non-linearities (turbulence).



[10] ARP, Stomberg et al., JHEP **07** (2025) 217, arXiv:2409.03651.

Primordial magnetic fields

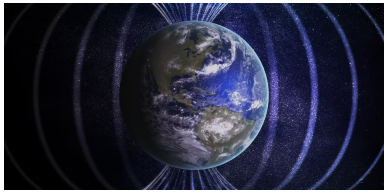
- Magnetic fields can either be produced at or present during cosmological phase transitions.
- The magnetic fields are strongly coupled to the primordial plasma and effectively produce vortical motion, inevitably leading to the development of MHD turbulence.[11]
- Present magnetic fields can be amplified by primordial turbulence via dynamo.[12]



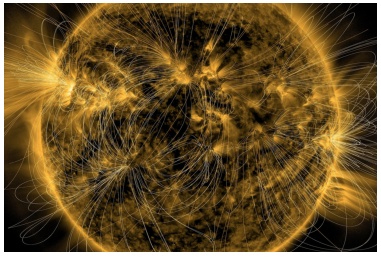
[11] J. Ahonen and K. Enqvist, *Phys. Lett. B* **382**, 40 (1996).

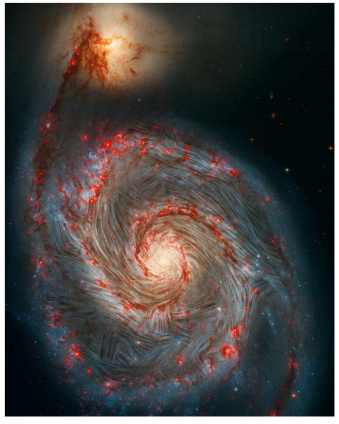
[12] A. Brandenburg *et al.* (incl. ARP), *Phys. Rev. Fluids* **4**, 024608 (2019).

Our magnetized Universe



$\sim 1\,\mathrm{G}$



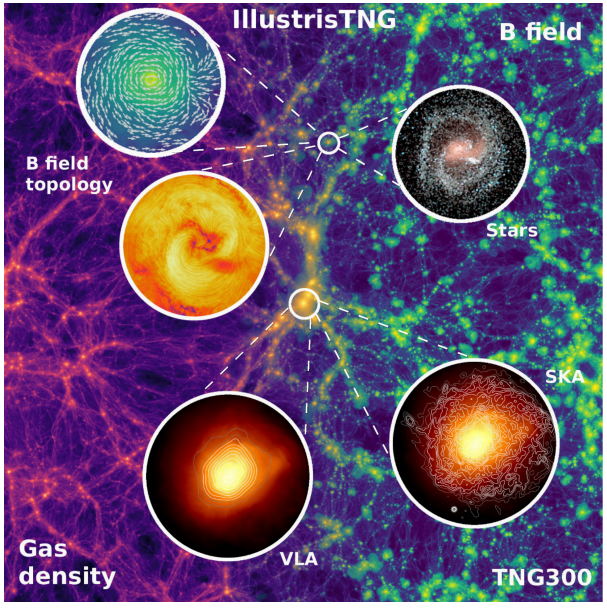


(FIR 145 μm)

(Radio 6 cm Fletcher et. al)

$\sim 10^{-5}\,\mathrm{G}$

Borlaff et al. (2021), SOFIA, M51



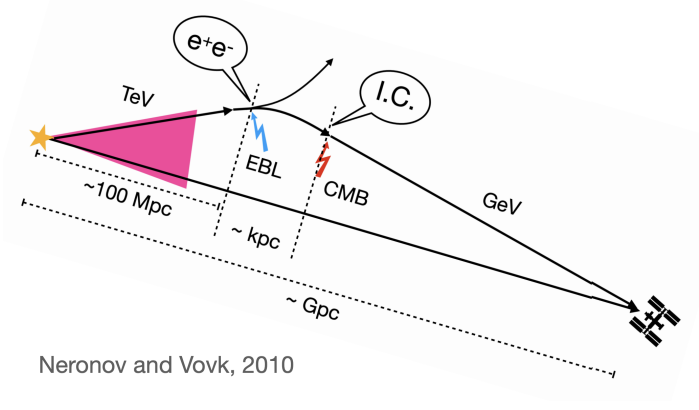
Marinacci et al. (2017), Illustris TNG

$\sim 10^{-9} - 10^{-6}\,\mathrm{G}$

- Dense regions in our Universe are strongly magnetized with $B \sim 10\mu$G in galaxies and clusters ($\sim$10 kpc). Radio and far-infrared (FIR) observations.
- Filaments also contain magnetic fields with $B \sim$10 nG (LOFAR observations of Faraday rotation measurements).

Intergalactic magnetic fields in voids of the LSS

- γ-ray observations from distant blazars by Fermi/MAGIC collaboration show a removal of power at GeV, providing evidence for an intergalactic magnetic field.
- Lower bound of $B \sim 10^{-16}$ G at Mpc scales.



Neronov and Vovk, 2010

Generation of primordial magnetic fields during inflation

- Magnetic fields could be amplified from quantum fluctuations during inflation.
- For inflationary magnetogenesis to be viable, conformal invariance needs to be broken,[13] with $\mathcal{L} \sim f(\phi) F_{\mu\nu} F^{\mu\nu}$ or $g(\phi) F_{\mu\nu} \tilde{F}^{\mu\nu}$. Otherwise, magnetic field perturbations would decay with the Universe expansion.
- After reheating, magnetic fields are strongly coupled to the primordial plasma and effectively produce vortical motion, inevitably leading to the development of MHD turbulence.[14]
- Alternatively, small seed magnetic fields can be amplified by primordial turbulence induced by phase transitions, e.g., via dynamo.[15]

[13] Turner & Widrow (1988), Ratra (1992), Gasperini *et al.* (1995).

[14] J. Ahonen and K. Enqvist, *Phys. Lett. B* **382**, 40 (1996).

[15] A. Brandenburg *et al.* (incl. ARP), *Phys. Rev. Fluids* **4**, 024608 (2019).

Generation of primordial magnetic fields during phase transitions

- Parity-violating processes during the electroweak phase transition ($T \sim 100$ GeV) are predicted by SM extensions that account for baryogenesis and can produce helical magnetic fields through sphaleron decay or B+L anomalies.[16]

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A} - i\frac{2\sin\theta_w}{gv^2}\boldsymbol{\nabla}\Phi^\dagger \times \boldsymbol{\nabla}\Phi$$

- Also after inflation, axion fields can amplify and produce magnetic fields.[17] For example, the QCD axion could oscillate and produce magnetic fields around the QCD scale ($T \sim 100$ MeV).[18]

$$\mathcal{L} \supset \frac{\phi}{f} F_{\mu\nu}\tilde{F}^{\mu\nu}$$

[16] T. Vachaspati, *Phys. Rev. B* **265**, 258 (1991), T. Vachaspati, *Phys. Rev. Lett.* **87**, 251302 (2001), J. M. Cornwall, *Phys. Rev. D* **56**, 6146 (1997).

[17] M. M. Forbes and A. R. Zhitnitsky, *Phys. Rev. Lett.* **85**, 5268 (2000).

[18] Miniati, Gregori, Reville & Sarkar (2018).

Generation of primordial magnetic fields from chiral anomalies

- Chiral magnetic effect in the early Universe at large temperatures $T > 80$ TeV can lead to the production of hypermagnetic fields.[19]
- Proposed chiral magnetic effect relevant at smaller scales[20] leading to a chiral plasma instability and development of magnetic fields that undergo inverse cascading.[21]
- Chiral magnetic effect also present for zero chiral chemical potential from local inhomogeneities.[22]
- Chiral magnetic effect can be activated from an active source of chiral chemical potential (e.g. associated to baryogenesis as the electroweak scale)[23]

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[19] M. Joyce and M. E. Shaposhnikov, *PRL* **79**, 1193 (1997)

[20] A. Boyarsky *et al. PRL* **108**, 031301 (2012).

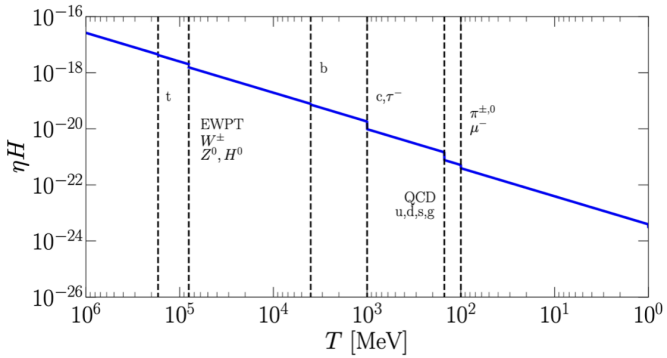
[21] I. Rogachevskii *et al. Astrophys. J. Lett* **845** (2017) L21, J. Schober *et al. Astrophy. J.* **858**, 124 (2018).

[22] J. Schober *et al. PRL* **132**, 6 (2024)

[23] M. Gurgenidze *et al.* (incl. ARP), arXiv:2512.09177 (2025).

MHD turbulence (large conductivity and small viscosity)

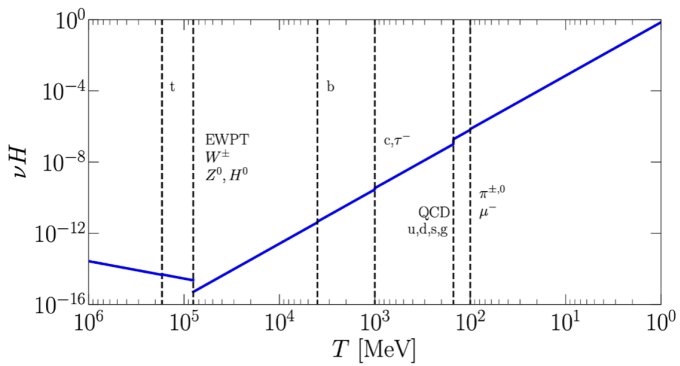
- High-conductivity (low magnetic diffusivity $\eta = 1/\sigma$) of the early universe leads to a high-coupling between magnetic and velocity fields.[24]



$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$$

$$\mathrm{R_m} = \frac{UL}{\eta}.$$

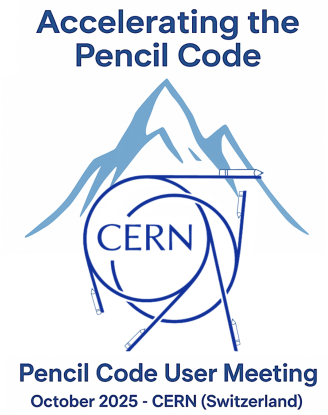
- Small viscosity of the early Universe plasma yields high Re number and turbulence



$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} - \nu \nabla^2 \mathbf{u} = -\frac{1}{\rho}\nabla p + \mathbf{g}$$

$$\mathrm{Re} = \frac{uL}{\nu}$$

[24] ARP, Midiri, JCAP **10** (2026) 029, arXiv:2501.05732.



Primordial magnetic fields and GWs from MHD turbulence

- During the radiation-dominated era, after they are produced, magnetic fields will decay following MHD turbulence.
- Direct numerical simulations using the Pencil Code[25] to solve:
  1. Relativistic MHD equations adapted for radiation-dominated era (after electroweak symmetry is broken).
  2. Gravitational waves equation.
- In general, large-scale simulations are necessary to solve the MHD nonlinearities (e.g., unequal-time correlators UETC and non-Gaussianities, which require simplifying assumptions in analytical studies).

[25] Pencil Code Collaboration, JOSS **6**, 2807 (2020), https://github.com/pencil-code/

Conservation laws for MHD turbulence

$$\nabla_\nu T^{\mu\nu} = 0, \quad \nabla_\nu F^{\mu\nu} = -J^\mu, \quad \nabla_\nu \mathcal{F}^{\mu\nu} = 0$$

In the limit of subrelativistic bulk flow:

$$\gamma^2 \sim 1 + (v/c)^2 + \mathcal{O}(v/c)^4$$

Relativistic MHD equations are reduced to[26]

$$\begin{aligned}
\frac{\partial \ln \rho}{\partial t} &= -\frac{4}{3}\left(\boldsymbol{\nabla}\cdot\boldsymbol{u} + \frac{1}{2}\boldsymbol{u}\cdot\boldsymbol{\nabla}\ln\rho\right) + \frac{1}{\rho}\left[\eta \boldsymbol{J}^2 - \boldsymbol{u}\cdot(\boldsymbol{J}\times\boldsymbol{B})\right], \\
\frac{D\boldsymbol{u}}{Dt} &= \frac{\mathbf{u}}{3}\left(\boldsymbol{\nabla}\cdot\boldsymbol{u} + \frac{1}{2}\boldsymbol{u}\cdot\boldsymbol{\nabla}\ln\rho\right) - \frac{\boldsymbol{u}}{\rho}\left[\eta J^2 + \frac{1}{2}\boldsymbol{u}\cdot(\boldsymbol{J}\times\boldsymbol{B})\right] \\
&\quad - \frac{1}{4}\boldsymbol{\nabla}\ln\rho + \frac{3}{4\rho}\boldsymbol{J}\times\boldsymbol{B} + \frac{2}{\rho}\boldsymbol{\nabla}\cdot(\rho\nu\boldsymbol{S}), \\
\frac{\partial \boldsymbol{B}}{\partial t} &= \boldsymbol{\nabla}\times(\boldsymbol{u}\times\boldsymbol{B} - \eta\boldsymbol{J}), \quad \boldsymbol{J} = \boldsymbol{\nabla}\times\boldsymbol{B} - \frac{\partial \boldsymbol{E}}{\partial t},
\end{aligned}$$

for a flat expanding universe with comoving and normalized $p = a^4 p_{\rm phys}, \rho = a^4 \rho_{\rm phys}, B_i = a^2 B_{i,{\rm phys}}, u_i$, and conformal time t ($\mathrm{d}t = a\mathrm{d}t_c$).

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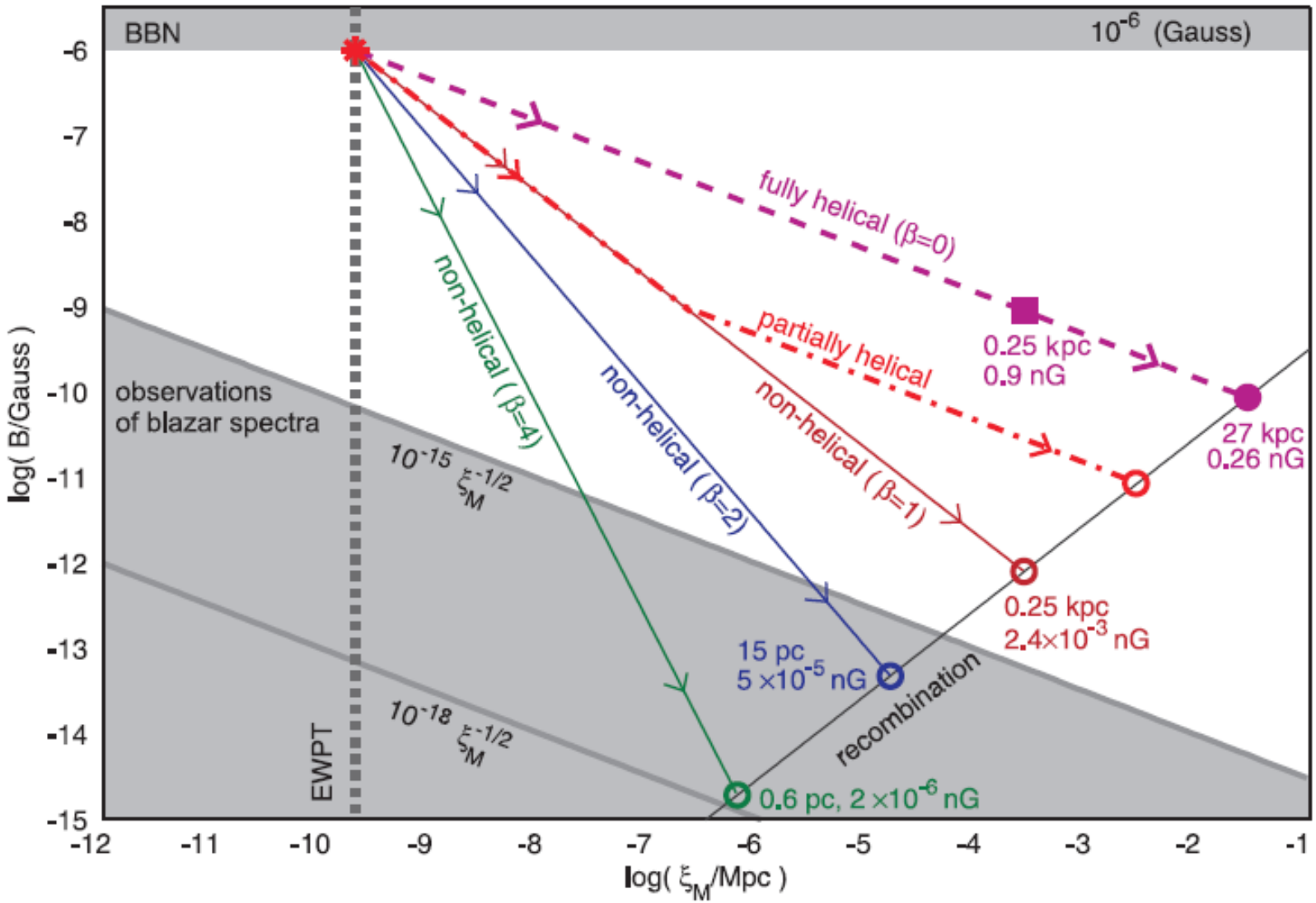
[26] A. Brandenburg, *et al.*, *Phys. Rev. D* **54**, 1291 (1996).
ARP, Midiri, *Relativistic Magnetohydrodynamics in the early Universe*, *JCAP* **06** (2026) 070, arXiv:2501.05732 (2025).

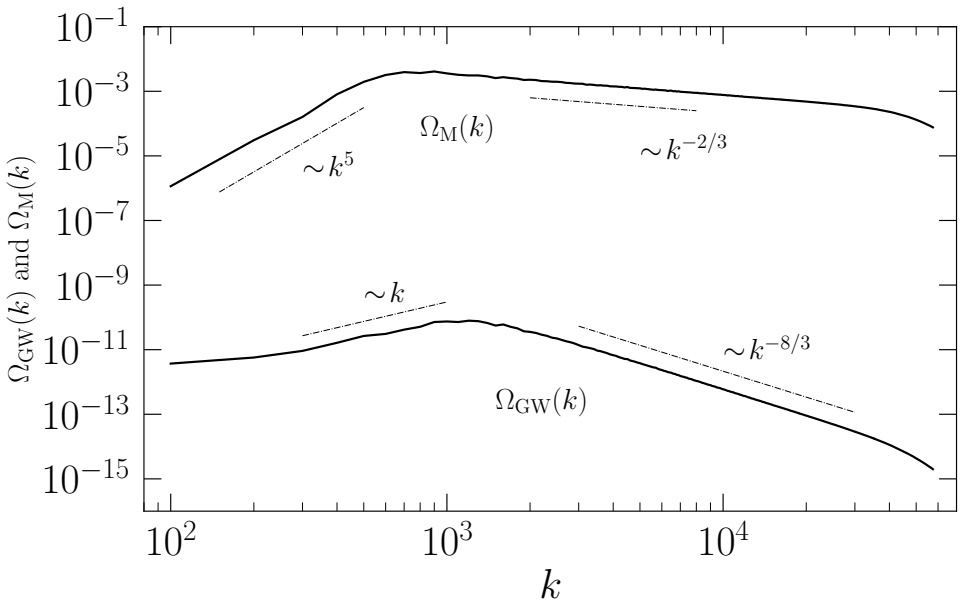
Evolution of magnetic strength and correlation length[27]



[27] A. Brandenburg *et al.* (incl. ARP), *Phys. Rev. D* **96**, 123528 (2017).

Numerical results for decaying MHD turbulence[28]

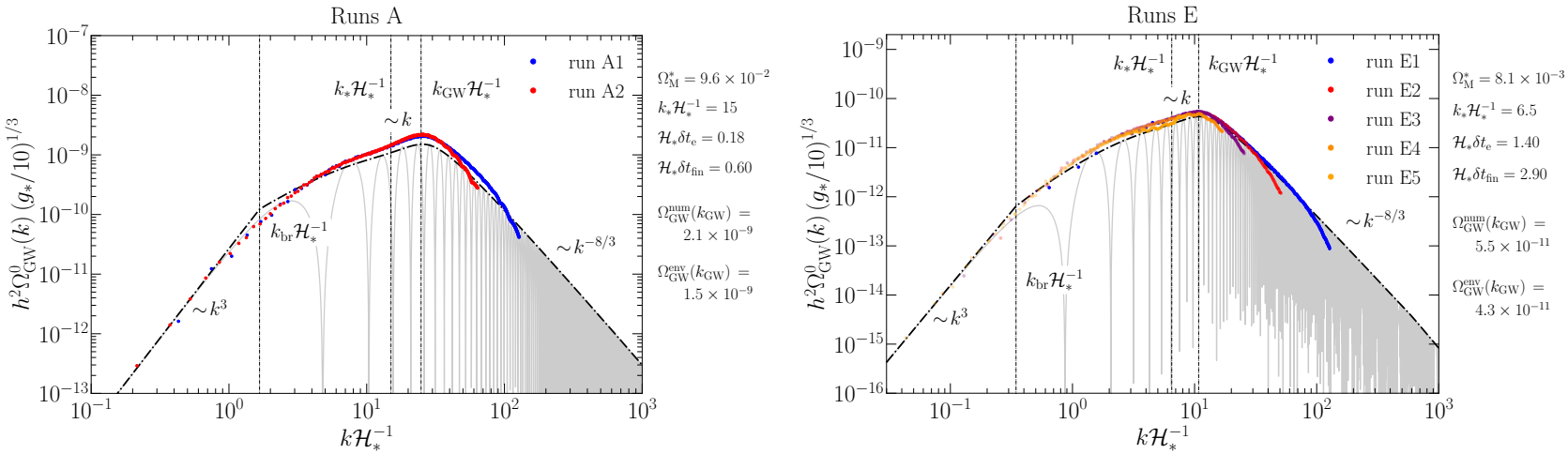
$1152^3, k_* = 2\pi \times 100, \Omega_{\rm M} \sim 10^{-2}, \sigma_{\rm M} = 1$



- **Characteristic k scaling in the subinertial range for the GW spectrum.**
- k^2 expected at scales $k < k_*$ and k^3 at $k < H_*$ according to the "top-hat" model (Caprini *et al.*, 2020).

[28] ARP *et al.* (Pittsburgh-gang dominated), *Phys. Rev. D* **102**, 083512 (2020).

Numerical results for decaying MHD turbulence[29]



run	$\Omega^*_{\rm M}$	$k_*\mathcal{H}_*^{-1}$	$\mathcal{H}_*\delta t_e$	$\mathcal{H}_*\delta t_{\rm fin}$	$\Omega^{\rm num}_{\rm GW}(k_{\rm GW})$	$[\Omega^{\rm env}_{\rm GW}/\Omega^{\rm num}_{\rm GW}](k_{\rm GW})$	n	$\mathcal{H}_*L$	$\mathcal{H}_*t_{\rm end}$	$\mathcal{H}_*\eta$
A1	9.6×10^{-2}	15	0.176	0.60	2.1×10^{-9}	1.357	768	6π	9	10^{-7}
A2	–	–	–	–	–	–	768	12π	9	10^{-6}
E1	8.1×10^{-3}	6.5	1.398	2.90	5.5×10^{-11}	1.184	512	4π	8	10^{-7}
E2	–	–	–	–	–	–	512	10π	18	10^{-7}
E3	–	–	–	–	–	–	512	20π	61	10^{-7}
E4	–	–	–	–	–	–	512	30π	114	10^{-7}
E5	–	–	–	–	–	–	512	60π	234	10^{-7}

[29] ARP *et al.*, *Phys. Rev. D* **105**, 123502 (2022).

Analytical CiT model for GWs from decaying turbulence

- Assumption: magnetic or velocity field evolution $\delta t_{\rm e} \sim 1/(u_* k_*)$ is slow compared to the GW dynamics ($\delta t_{\rm GW} \sim 1/k$) at all $k \gtrsim u_* k_*$.
- We can derive an analytical expression for nonhelical fields of the envelope of the oscillations[30] of $\Omega_{\rm GW}(k)$.

$$\Omega_{\rm GW}(k,t_{\rm fin}) \approx 3 \left(\frac{k}{k_*}\right)^3 {\Omega_{\rm M}^*}^2 \frac{\mathcal{C}(\alpha)}{\mathcal{A}^2(\alpha)} \, p_\Pi\left(\frac{k}{k_*}\right) \times \begin{cases} \ln^2[1+\mathcal{H}_* \delta t_{\rm fin}] & \text{if } k\,\delta t_{\rm fin} < 1, \\ \ln^2[1+(k/\mathcal{H}_*)^{-1}] & \text{if } k\,\delta t_{\rm fin} \geq 1. \end{cases}$$

- p_Π is the anisotropic stress spectrum and depends on spectral shape, can be approximated for a von Kárman spectrum as[31]

$$p_\Pi(k/k_*) \simeq \left[1+\left(\frac{k}{2.2k_*}\right)^{2.15}\right]^{-11/(3\times 2.15)}$$

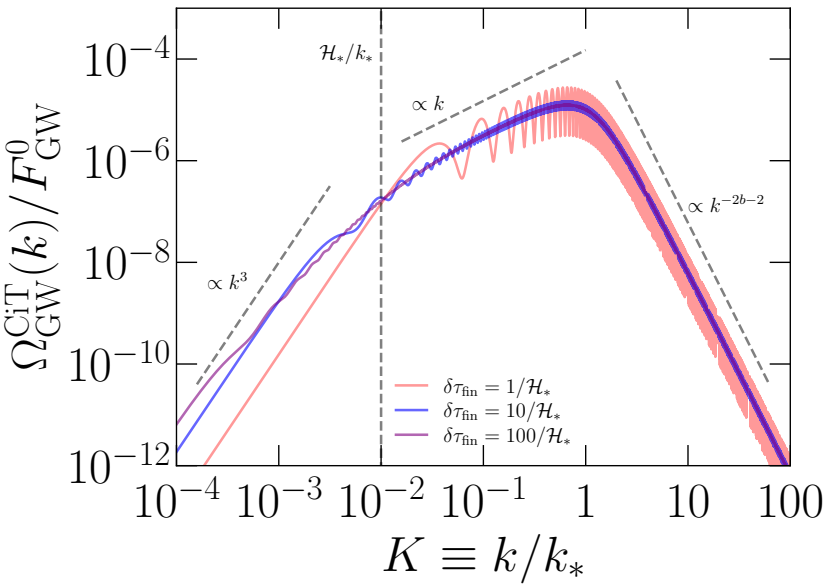
[30] ARP *et al.*, *Phys. Rev. D* **105**, 123502 (2022); ARP, Midiri, Salomé, Caprini, in preparation.

[31] ARP *et al.*, arXiv:2307.10744 (2023).

Constant-in-Time (CiT) model (arXiv:2611.xxzz)

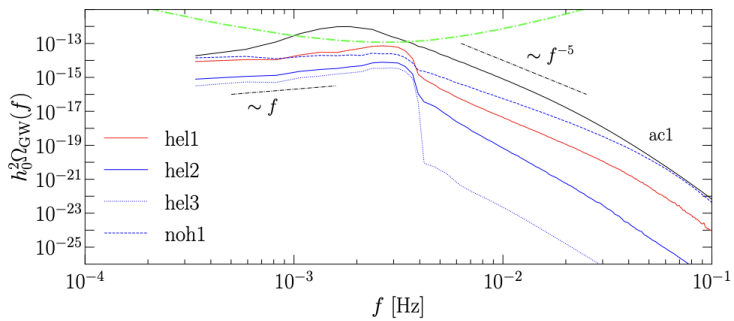
[PRELIMINARY work with A. Midiri, M. Salomé, C. Caprini]

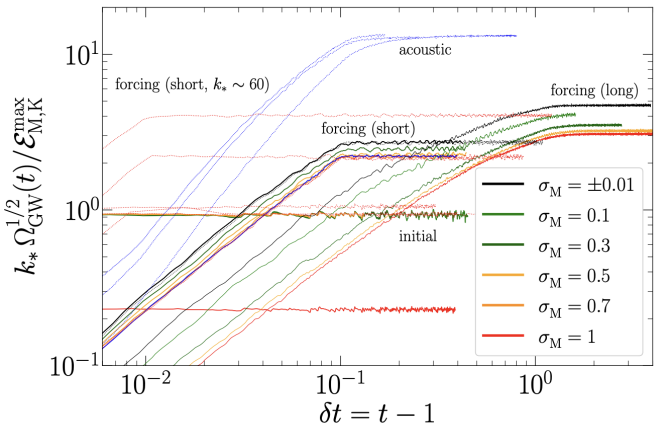
- Applicable to generic slowly-decaying sources
- Extension to include decay of the source impacts IR part of the spectrum
- Used to compute circular polarization, compatible with numerical results.
- Applicable to generic vector and scalar fields sourcing GWs.
- Application to sound waves when $c_s \ll 1$ and to turbulence when $u_{dc} \ll 1$.



Acoustic vs vortical sourcing

- Numerical simulations show that acoustic forcing is more efficient than vortical in producing GWs [32]





- Counterintuitive since vortical motion contains more energy in the tensor-mode of stresses[33].

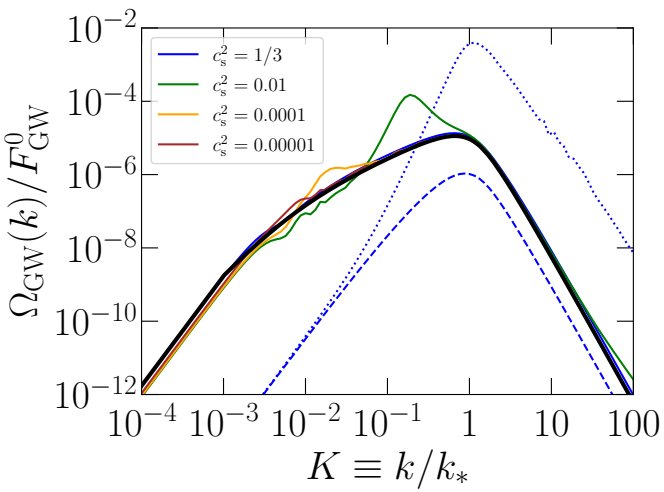
[32] ARP *et al.*, *Phys.Rev.D* **102** (2020) 8, 083512
ARP *et al.*, *JCAP* **04** (2022) 04, 019.

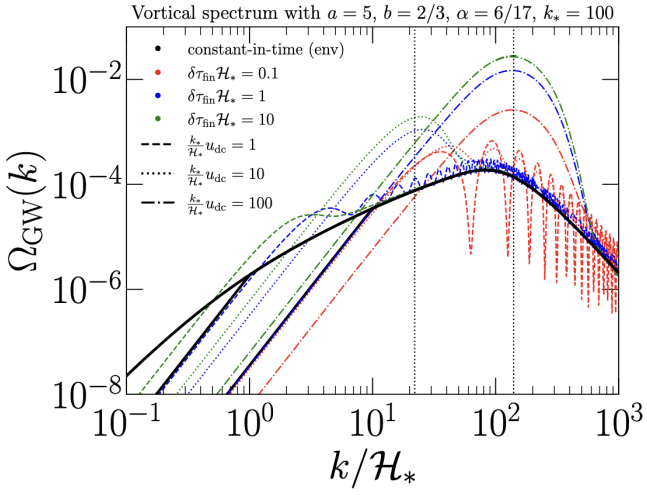
[33] Mandal *et al.* (incl. ARP), *Class. Quant. Grav.* **38** (2021) 14, 145002

Acoustic vs vortical sourcing

[PRELIMINARY work with A. Midiri, M. Salomé, C. Caprini]

- Key point to solve this question is the applicability (and non-applicability) of the CiT model to sound waves and turbulence.
- The unequal-time correlator for sound waves and turbulence plays a significant role at relativistic $c_{\rm s}$ and $u_{\rm dc}$.



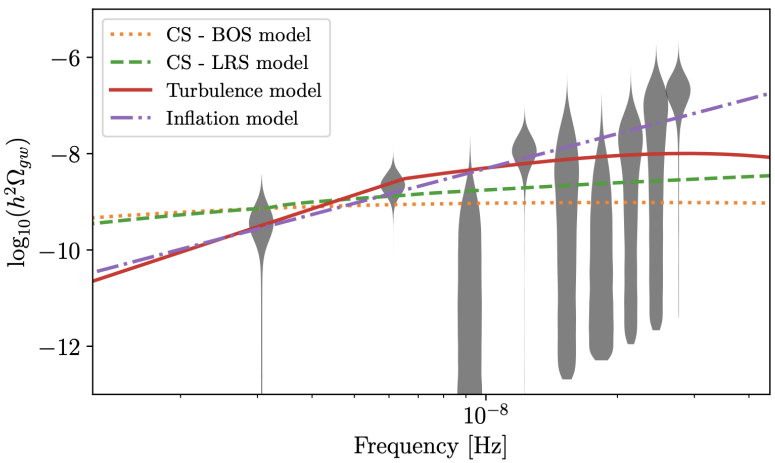


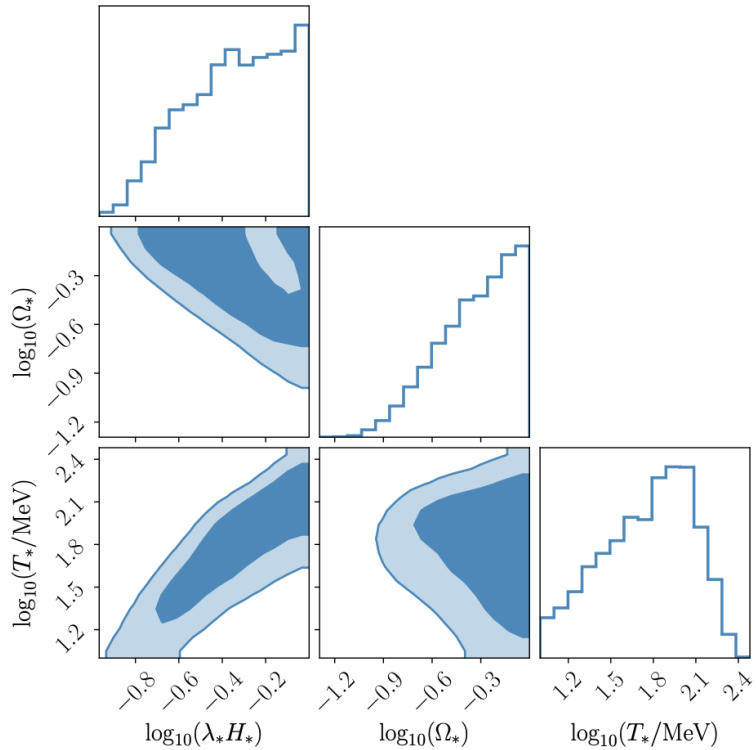
CosmoGW review paper and official release (soon)

Documentation available on Read the Docs, source code on GitHub.

- Routines to generate GW backgrounds from sound waves and turbulence associated to phase transitions
- Reading routines from numerical simulations (e.g., Pencil Code)
- Efficient numerical integration techniques
- Cosmology calculations
- Interferometry, detector sensitivities

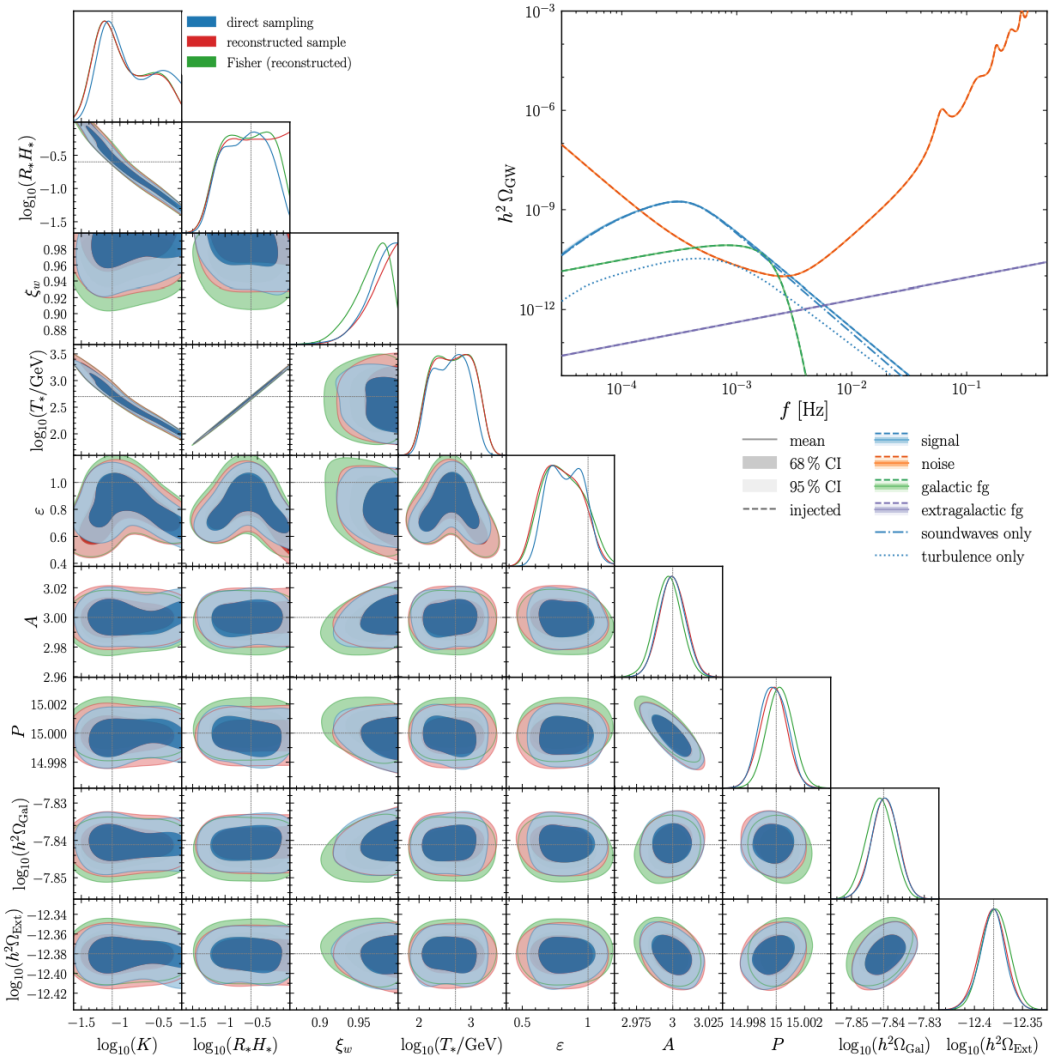
Primordial magnetic fields constraints with PTA[34]





[34] [EPTA/InPTA collab.] (incl. ARP), arXiv:2306.16227 (2023).

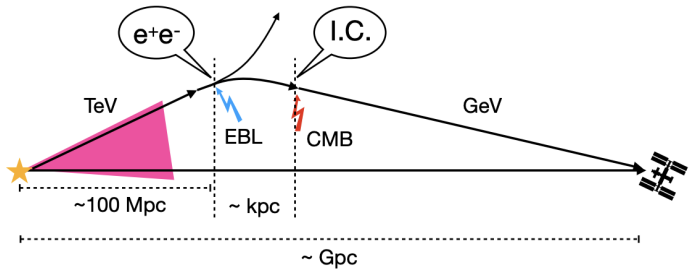
Parameter reconstruction of GW signal for sound waves and turbulence from a PT with LISA[35]



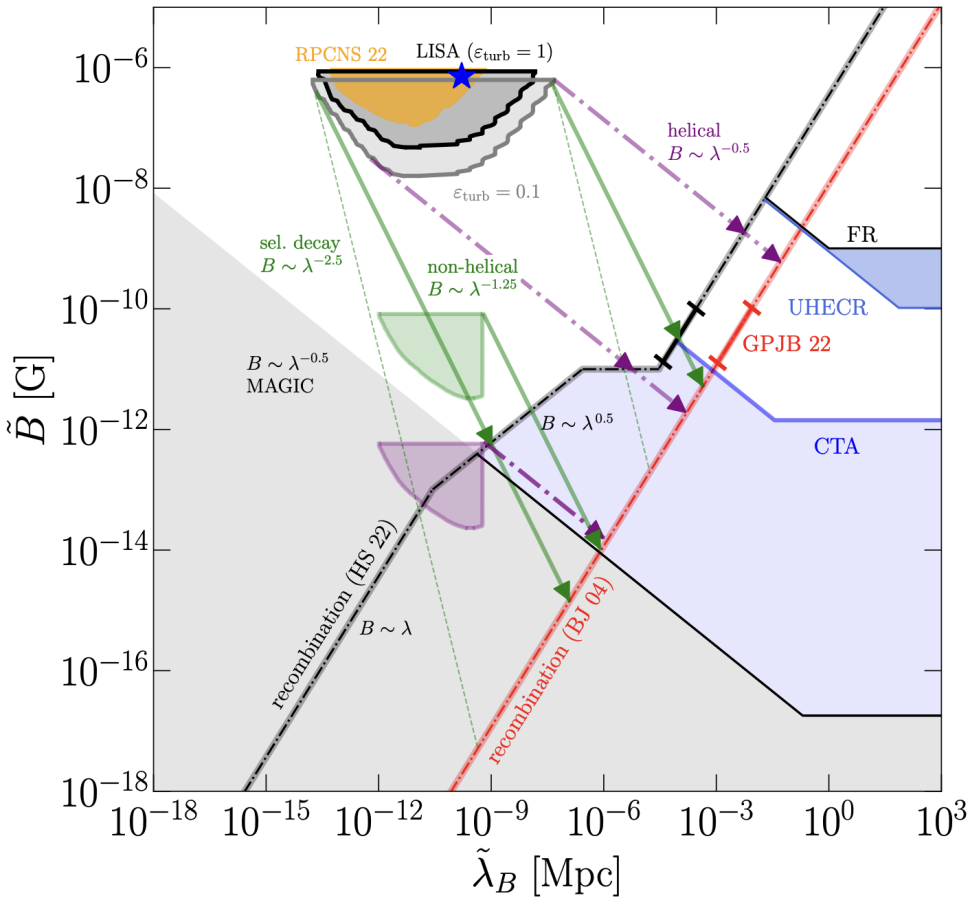
[35] [LISA CosWG] (incl. ARP), arXiv:2403.03723 (2024).

Multi-messenger studies of primordial magnetic fields[30]

- Primordial magnetic fields would evolve through the history of the universe up to the present time and could explain the lower bounds in cosmic voids derived by the Fermi collaboration.[31]



- Maximum amplitude of primordial magnetic fields is constrained by the big bang nucleosynthesis.[32]
- Additional constraints from CMB, Faraday Rotation, ultra-high energy cosmic rays (UHECR).



[30] ARP *et al.*, arXiv:2307.10744 (2023).

[31] A. Neronov and I. Vovk, *Science* **328**, 73 (2010).

[32] V. F. Shvartsman, *Pisma Zh. Eksp. Teor. Fiz.* **9**, 315 (1969).

(How) can AI help us?

- Can we find **universal laws** in decaying MHD turbulence? Can AI help us find local or global **invariants** in simulated data that determine the decay of acoustic and vortical turbulence, including primordial magnetic fields?
- Can we accelerate and improve the efficiency of **numerical simulations with AI agents?** Can we develop adaptive mesh refinement, adaptive numerical schemes on-line using machine learning?
- *MHD simulations are subjected to constraints (e.g. Gauss law) that are not always satisfied numerically. Can we use AI to apply* **projections to 3D fields** *avoiding expensive Fourier transforms?*
- Non-linear dynamics is extremely sensitive to small-scale dynamics. Can we use AI to **train subgrid models** that infer information on large-scale MHD perturbations?
- Can we use **simulation based-inference** to produce templates of GWBs avoiding computing them numerically for every parameter characterizing early Universe physics?
- *Can we* **infer unequal-time correlators** *of the source from the resulting GW background?*

Thank You!

alberto.roperpol@unige.ch

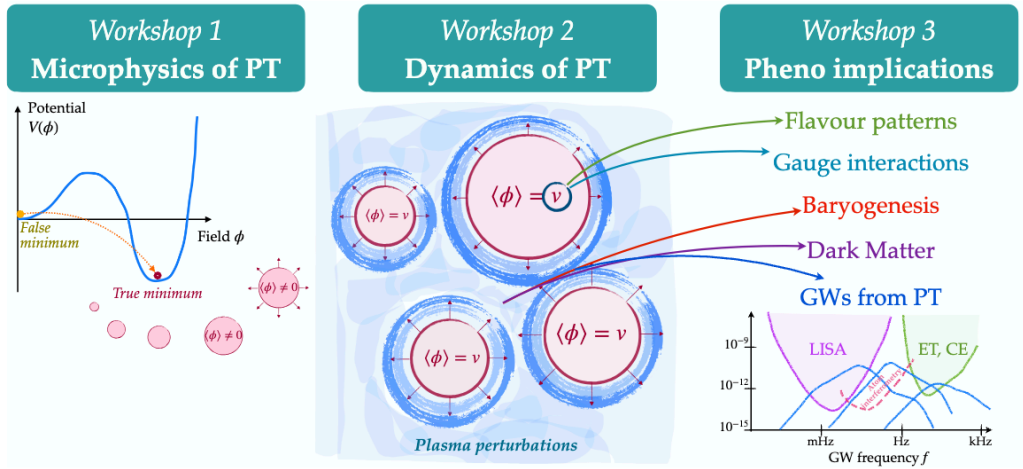
github.com/CosmoGW/CosmoGW

albertoroper.github.io

Exploring Cosmological Phase Transitions with Gravitational Waves

Scientific program at Nordita: 3 weeks + 1-week mini school

NORDITA



Save the date! June 7 - July 2, 2027

Co-organized with

M. Nemevšek, D. Racco, P. Schicho, B. Świeżewska.